

Task 3 — Tool Design Report

FairCredit Jurisdiction Navigator

A Jurisdiction-Aware AI Credit Decisioning Compliance Tool for American Express

1. Executive Summary

For Task 3, I designed and partially implemented FairCredit Jurisdiction Navigator, a prototype RegTech tool for American Express. The tool is built around a simple but important idea: the same AI-supported credit decision should not be interpreted in the same way across different jurisdictions.

The selected use case is AI-supported credit card approval. In this prototype, a machine learning model estimates whether a synthetic applicant is likely to repay well over the next 12 months. The model then produces an operational decision: Approved, Manual Review, or Declined. The compliance layer sits above this model and interprets the output differently depending on whether the active jurisdiction is the United States or the European Union.

The US mode focuses on individual adverse action explainability. This reflects the US regulatory concern that creditors using AI or complex credit models must still provide specific and accurate reasons when taking adverse action against consumers (Consumer Financial Protection Bureau [CFPB], 2023).

The EU mode focuses on system-level AI governance, fairness monitoring, model drift, and human oversight. This reflects the EU AI Act's treatment of AI systems used for creditworthiness assessment or credit scoring as high-risk AI systems (European Union, 2024).

The prototype uses 2,000 synthetic credit application records, not real American Express customer data. A logistic regression model is trained on credit risk variables such as income, debt-to-income ratio, credit history length, missed payments, credit utilisation, employment status, and spending volatility. A synthetic group proxy is excluded from model training and used only for post-model fairness monitoring. This design choice follows the values position from Task 2: the tool should measure genuine risk and not merely document compliance.

In the test sample of 600 scored applications, the model approved 500 applications, sent 50 to manual review, and declined 50. In EU fairness monitoring, Group B had the lowest approval rate at 73.96%, compared with 88.15% for Group A. The approval-rate disparity ratio was 0.839, which does not breach the prototype threshold of 0.80, but still suggests a monitoring signal. The stronger governance signal came from drift monitoring: the Population Stability Index for debt-to-income was 0.237, above the prototype threshold of 0.20. This means that EU mode would trigger human model governance review.

Overall, the prototype demonstrates a realistic jurisdiction-aware compliance logic: US mode asks whether the firm can explain a specific declined decision; EU mode asks whether the AI credit system is being properly

governed and monitored at portfolio level.

2. Product Context and Problem Definition

American Express is a suitable target for this tool because its business is closely connected to card issuing, consumer credit, payments, and risk management. American Express National Bank is listed by the Office of the Comptroller of the Currency as a national bank, which gives the US side of the project a real regulated entity context. (OCC.gov)

The problem is not that American Express lacks credit models. The problem is that AI-supported credit models may create different compliance obligations in different jurisdictions. A model score is not self-explanatory. It does not automatically tell a compliance officer whether the decision is explainable, fair, stable, or properly governed.

A memo or spreadsheet can describe regulatory differences, but it cannot continuously apply jurisdiction-specific rules to actual model outputs. This is where FairCredit Jurisdiction Navigator adds value. It connects: model outputs; jurisdiction-specific regulatory logic; individual explanations; group-level fairness monitoring; drift detection; human review triggers.

In practical terms, the tool does not replace the credit model. It sits above the model as a compliance and governance layer.

3. Design Objectives

The prototype has four main objectives.

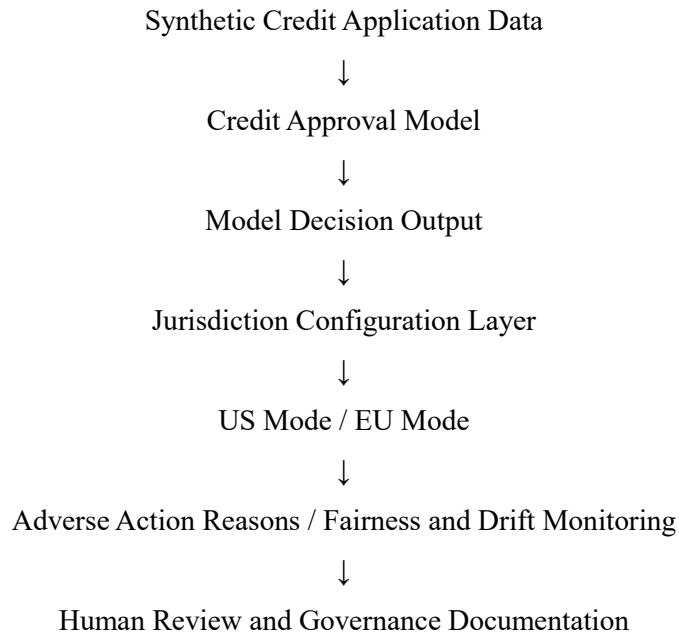
Objective	Design Response
Show jurisdictional divergence	US and EU modes apply different compliance logic to the same model output
Support explainability	US mode generates applicant-level adverse action reasons
Monitor system-level risk	EU mode calculates fairness and drift indicators
Preserve human judgement	Borderline cases, fairness signals, and drift alerts trigger human review

I deliberately kept the prototype simple. I used logistic regression rather than a more complex black-box model because the aim is not to maximise predictive performance. The aim is to demonstrate how a jurisdiction-aware compliance layer can work in a credit decisioning environment.

This is also consistent with the NIST AI Risk Management Framework, which emphasises governance, mapping, measurement, and management of AI risks across the AI lifecycle rather than treating AI risk as a one-time documentation exercise (National Institute of Standards and Technology [NIST], 2023).

4. System Architecture

The prototype architecture is intentionally modular:



4.1 Data Layer

The data layer contains synthetic credit application records. Each record includes applicant-level variables that are plausible in a credit card approval setting.

Main variables include:

Variable	Purpose
annual_income	Measures repayment capacity
debt_to_income	Captures debt burden
credit_history_years	Captures credit record depth
missed_payments_12m	Captures recent repayment problems
credit_utilization	Captures current credit pressure
employment_status	Captures income stability
spending_volatility	Captures behavioural risk
requested_credit_limit	Captures exposure requested
synthetic_group_proxy	Used only for fairness monitoring
good_repayment_12m	Synthetic target label
model_decision	Approved / Manual Review / Declined

The important design choice is that `synthetic_group_proxy` is not used for model training. It is only used for post-model fairness analysis. This avoids directly encoding group identity into the credit model while still allowing the tool to test whether outcomes differ across groups.

4.2 Model Layer

The model layer uses logistic regression to estimate the probability that an applicant will have good repayment behaviour over the next 12 months. The model output is then converted into an operational decision:

Probability Range	Model Decision
≥ 0.62	Approved
0.48–0.62	Manual Review
< 0.48	Declined

These thresholds are illustrative. They are not intended to represent American Express’s actual credit policy.

4.3 Compliance Layer

The compliance layer is the main contribution of the prototype. It does not simply display model scores. It interprets the same output differently depending on the active jurisdiction.

5. Jurisdiction Configuration Layer

The jurisdiction configuration layer is the core of the tool. In the prototype, it is implemented as a simple Python dictionary. In a real system, it would need to be stored in a versioned rule engine with owners, effective dates, legal review, and audit logs.

Configuration Area	US Mode	EU Mode
Regulatory focus	Adverse action explainability and fair lending	High-risk AI governance and bias monitoring
Individual explanations	Required	Required but not sufficient
Fairness dashboard	Not central in prototype	Required
Drift monitoring	Not central in prototype	Required
Human review	Borderline or declined cases	Fairness, drift, and borderline cases
Main question	Can we explain this decision?	Is the system governed and monitored properly?

This design directly reflects the difference between the two jurisdictions. In the United States, the CFPB has emphasised that creditors using AI or complex algorithms must still provide specific and accurate adverse action reasons (CFPB, 2023). In the European Union, the AI Act takes a broader governance approach to creditworthiness-related AI systems by treating them as high-risk AI systems (European Union, 2024).

6. US Mode: Adverse Action Explanation

The US mode focuses on applicant-level explainability. If the model declines an applicant or sends the application to manual review, the tool generates adverse action reasons based on the applicant’s risk factors.

The prototype uses a transparent rule-based explanation layer. For example, it checks whether the applicant has: high debt-to-income ratio; limited credit history; recent missed payments; high credit utilisation; high spending volatility; income below an internal risk threshold; requested credit limit that is high relative to income.

A typical US mode output looks like this:

Jurisdiction: US

Regulatory focus: Adverse action explainability and fair lending

Model decision: Declined / Manual Review

Adverse action reasons:

1. High debt-to-income ratio
2. Limited credit history
3. Recent missed payments

Compliance interpretation:

If the application is declined, the institution should provide specific and accurate reasons.

A generic explanation such as "low model score" would not be sufficient.

The key point is that the tool does not treat "low model score" as an acceptable explanation. This matters because a consumer-facing adverse action explanation should be linked to understandable factors, not hidden behind the existence of an algorithm.

This part of the tool is deliberately simple. A future version could add SHAP or LIME for more detailed model explainability. However, I kept the prototype rule-based because the explanations are transparent and easier to audit.

7. EU Mode: Fairness and High-Risk AI Monitoring

The EU mode evaluates the model at portfolio level. It does not stop at individual explanations. Instead, it asks whether the credit decisioning system shows signs of unfair or unstable behaviour.

The EU mode calculates:

Metric	Purpose
Approval rate by group	Checks whether one group is approved much less often
Approval-rate disparity ratio	Compares lowest and highest group approval rates
False positive rate by group	Checks whether risky applicants are wrongly approved at different rates
False negative rate by group	Checks whether good applicants are wrongly rejected at different rates
Average model probability by group	Checks whether the model scores groups differently
PSI drift	Checks whether the applicant population has shifted

In the prototype, the approval-rate disparity ratio is defined as: lowest group approval rate / highest group approval rate. A ratio below 0.80 triggers a fairness alert.

7.1 Fairness Monitoring Output

The fairness output from the test sample is:

Synthetic Group	Records	Approval Rate	Average Model Probability
Group A	346	88.15%	0.841
Group B	169	73.96%	0.749
Group C	85	82.35%	0.814

The lowest approval rate is Group B at 73.96%, while the highest approval rate is Group A at 88.15%. The approval-rate disparity ratio is: $0.7396 / 0.8815 = 0.839$

Since 0.839 is above the prototype threshold of 0.80, the tool does not trigger a hard fairness alert. However, I would not describe this as a clean pass. Group B still has a meaningfully lower approval rate and lower average model probability. In a real deployment, I would treat this as a monitoring signal and continue investigation before assuming the model is fair.

This result is useful because it shows the difference between threshold-based compliance and genuine risk awareness. A model can pass a threshold but still show patterns worth monitoring.

7.2 Error Rate Output

The prototype also calculates false positive and false negative rates by group:

Group	False Positive Rate	False Negative Rate
Group A	86.44%	4.88%
Group B	64.58%	5.79%
Group C	61.54%	4.17%

The false positive rates are relatively high. I interpret this as a limitation of the prototype, not as a production-ready risk result. The synthetic dataset contains a high share of applicants with acceptable repayment profiles, and the model threshold is relatively approval-friendly. In a real deployment, credit risk and model risk teams would need to calibrate thresholds against the institution's actual risk appetite.

This is an important honesty point. The prototype demonstrates compliance mechanics, not a validated credit risk model.

8. Drift Monitoring and Governance Review

The prototype also simulates a shifted applicant population and calculates Population Stability Index, or PSI. PSI is used here as a simple indicator of whether the current applicant population differs from the baseline model population.

The interpretation in the prototype is:

PSI Range	Interpretation
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PSI Range	Interpretation
< 0.10	Limited shift
0.10–0.20	Moderate shift
> 0.20	Material shift requiring review

The PSI results are:

Variable	PSI	Review Required
annual_income	0.032	No
debt_to_income	0.237	Yes
credit_utilization	0.178	No, but moderate
missed_payments_12m	0.092	No

The strongest governance signal is debt-to-income, with a PSI of 0.237, which exceeds the prototype review threshold of 0.20. This means that EU mode would trigger human model governance review.

This is one of the most important findings from the prototype. The model did not breach the hard fairness threshold, but it did show a material drift warning in a key credit risk variable. This shows why a jurisdiction-aware tool should not only detect bias after harm appears. It should also detect when the applicant population changes enough to justify model review.

For example, if debt-to-income levels rise across the applicant population, the model may become less reliable if it was trained on a more stable baseline population. This could lead to unexpected approval behaviour, increased credit risk, or unfair treatment of applicants whose risk is being assessed using outdated assumptions.

9. Human Judgement Boundary

The tool is designed to support human judgement, not replace it. This is especially important because credit decisions involve legal, ethical, and business judgement.

The prototype routes the following situations to human review:

Trigger	Reason for Human Review
Manual Review decision	Applicant is near the model threshold
Declined decision in US mode	Adverse action reasons must be checked for clarity and accuracy
Approval-rate disparity below threshold	Possible group-level fairness issue
Material PSI drift	Model population has shifted
Unusual error rate differences	Model may behave unevenly across groups
Jurisdictional uncertainty	Compliance logic may require legal interpretation

This boundary is necessary because the tool cannot decide whether a specific explanation is legally sufficient in every case. It also cannot determine whether a group-level disparity is justified by legitimate credit risk differences. Those decisions require human compliance, model risk, legal, and credit risk review.

The tool's role is therefore to classify, explain, flag, and document. It should not make the final legal or ethical decision.

10. Failure Mode Analysis

A strong RegTech tool must be honest about where it can fail. I identified five main failure modes.

Failure Mode	Example	Main Harm	Mitigation
False positive alert	A compliant model is flagged as risky	Operational delay and review cost	Calibrate thresholds and allow reviewer notes
False negative alert	A biased pattern is not detected	Consumers may be harmed without warning	Use multiple fairness and drift metrics
Model drift	Applicant population changes over time	Model becomes less reliable	PSI monitoring and periodic recalibration
Jurisdictional misconfiguration	EU portfolio assessed under US-only logic	Wrong compliance obligations applied	Versioned jurisdiction rules and audit logs
Over-trust in tool output	Staff treat tool output as final legal answer	Human judgement becomes weakened	Clear disclaimers and mandatory review points

The most serious risk is a false negative. If the tool fails to detect unfair outcomes or vague adverse action reasons, consumers may be denied credit without meaningful explanation. American Express may also face legal, regulatory, and reputational consequences.

The most central risk for this project is jurisdictional misconfiguration. If EU cases are assessed only through US adverse action logic, the tool may generate explanations but fail to trigger broader EU-style high-risk AI governance checks. That would defeat the purpose of the tool.

11. What the Tool Does Not Do

This section is important because the tool should be honest about its boundaries.

FairCredit Jurisdiction Navigator does not:

1. use real American Express customer data;
2. reproduce American Express's actual credit scoring model;
3. provide legal advice;
4. make final approval or rejection decisions;
5. guarantee compliance with all US or EU rules;
6. cover all EU member state differences;
7. handle AML/KYC or fraud monitoring;
8. replace human compliance or model risk review.

These limitations do not make the prototype useless. They define its proper purpose. The prototype is a proof of concept showing how jurisdiction-aware regulatory logic can be connected to AI credit decision outputs.

If I had more time and data, I would add: SHAP-based local explanations; more realistic credit bureau variables; more granular EU member state configuration; dashboard interface; legal rule versioning; reviewer workflow and audit log; stress testing under macroeconomic scenarios; integration with model card generation.

12. Why This Tool Is Better Than a Memo or Spreadsheet

A memo can explain the difference between the US and EU regulatory approaches, but it cannot apply those differences to live model outputs. A spreadsheet can calculate approval rates, but it does not provide jurisdiction-specific workflow, adverse action explanations, drift alerts, and governance documentation in one place.

The value of FairCredit Jurisdiction Navigator is that it connects:

model output → jurisdiction rule → compliance interpretation → human review

This makes the tool more practical than static documentation. It also makes regulatory divergence visible in the actual decision process.

13. Conclusion

FairCredit Jurisdiction Navigator demonstrates the main idea behind my Task 3 design: a RegTech tool should not treat AI credit decisions as having the same compliance meaning everywhere.

In the prototype, the same credit model output is interpreted differently under US and EU modes. The US mode focuses on individual adverse action explanations. The EU mode focuses on high-risk AI governance, fairness monitoring, drift detection, and human oversight.

The prototype results support this distinction. The model produced operational decisions across 600 test applications. The fairness monitoring output showed that Group B had a lower approval rate than Group A, although the approval-rate disparity ratio did not breach the hard threshold. The drift monitoring output produced a stronger governance signal, with debt-to-income PSI exceeding the review threshold. This means the EU mode would trigger human review even when the fairness threshold is not formally breached.

My final view is that this tool is not a replacement for legal judgement, credit risk expertise, or model validation. Its value is that it helps make the right questions visible at the right time. In US mode, the key question is: Can this declined applicant receive a specific and accurate explanation? In EU mode, the key question is: Is this high-risk AI credit system being properly monitored and governed?

That difference is the reason the tool needs to be jurisdiction-aware.

References List:

1. Consumer Financial Protection Bureau. (2023, September 19). CFPB issues guidance on credit denials by lenders using artificial intelligence.
<https://www.consumerfinance.gov/about-us/newsroom/cfpb-issues-guidance-on-credit-denials-by-lenders-using-artificial-intelligence/>
2. European Union. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence. Official Journal of the European Union.
<https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
3. National Institute of Standards and Technology. (2023). Artificial intelligence risk management framework (AI RMF 1.0). U.S. Department of Commerce. <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>
4. Office of the Comptroller of the Currency. (n.d.). American Express National Bank. Financial Institution Search. Retrieved May 24, 2026, from <https://www OCC.treas.gov/institution-search/>